

Week 3 — SQL & Excel for Data Analytics

Practical Assignment Report

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Program: B.Tech Computer Engineering

Task 1 — SELECT

SELECT *, selected columns and aliases.

Task 1 — SELECT Statement Output

Student_Name	Final_Marks
Aarav Sharma	86
Diya Patel	91
Rohan Mehta	78
Anaya Shah	88
Vivaan Desai	74
Isha Joshi	95
Kabir Singh	69
Meera Patel	84
Arjun Kumar	89
Sara Khan	76

Task 2 — WHERE, ORDER BY & Aggregates

Filtering, sorting and COUNT/SUM/AVG/MIN/MAX.

Task 2 — Aggregate Functions Output

Total_Students	Total_Marks	Average_Marks	Minimum_Marks	Maximum_Marks
10	830	83.0	69	95

Task 3 — GROUP BY & HAVING

Department grouping and HAVING filter.

Task 3 — GROUP BY & HAVING Output

Department	Student_Count	Average_Marks	Highest_Marks
Computer	3	86.33	95
Electronics	2	86.0	88
IT	3	84.67	91

Task 4 — SQL Joins

INNER JOIN output; SQL file also contains LEFT and RIGHT JOIN.

Task 4 — INNER JOIN Output

Student_ID	Name	Department	HOD	Location
101	Aarav Sharma	Computer	Dr. Mehta	Block A
102	Diya Patel	IT	Dr. Shah	Block B
103	Rohan Mehta	Computer	Dr. Mehta	Block A
104	Anaya Shah	Electronics	Dr. Patel	Block C
105	Vivaan Desai	IT	Dr. Shah	Block B
106	Isha Joshi	Computer	Dr. Mehta	Block A
107	Kabir Singh	Mechanical	Dr. Singh	Block D
108	Meera Patel	Electronics	Dr. Patel	Block C
109	Arjun Kumar	IT	Dr. Shah	Block B
110	Sara Khan	Mechanical	Dr. Singh	Block D

Task 5 — Subqueries

Employees above average salary; second product-sales subquery is also included.

Task 5 — Subquery Output

Employee_Name	Department	Salary
Nisha	IT	75000
Vikas	Finance	72000
Amit	IT	68000
Riya	Finance	64000
Neha	IT	61000

Task 6 — Excel Formatting

Sales data prepared for formatting, sorting and filtering.

Task 6 — Excel Data Formatting / Sorting / Filtering

Sale_ID	Product	Category	Quantity	Unit_Price	Total_Sales
1001	Laptop	Electronics	2	55000	110000
1002	Mouse	Accessories	8	700	5600
1003	Keyboard	Accessories	5	1200	6000
1004	Monitor	Electronics	3	14000	42000
1005	Laptop	Electronics	1	55000	55000
1006	Headphones	Accessories	6	2500	15000
1007	Printer	Office	2	12000	24000
1008	Monitor	Electronics	4	14000	56000
1009	Chair	Office	5	6500	32500
1010	Laptop	Electronics	3	55000	165000

Task 7 — Excel Functions

IF and conditional-formatting evidence; workbook also contains COUNTIF and SUMIF.

Task 7 — IF / Conditional Formatting Evidence

Sale_ID	Product	Category	Total_Sales	IF Result
1001	Laptop	Electronics	110000	High
1002	Mouse	Accessories	5600	Normal
1003	Keyboard	Accessories	6000	Normal
1004	Monitor	Electronics	42000	Normal
1005	Laptop	Electronics	55000	High
1006	Headphones	Accessories	15000	Normal
1007	Printer	Office	24000	Normal
1008	Monitor	Electronics	56000	High
1009	Chair	Office	32500	Normal
1010	Laptop	Electronics	165000	High
1011	Mouse	Accessories	7000	Normal
1012	Desk	Office	18000	Normal

Task 8 — VLOOKUP & XLOOKUP

Related product table prepared for both lookup methods.

Task 8 — VLOOKUP / XLOOKUP Source Table

Product	Category	Unit_Price
Laptop	Electronics	55000
Mouse	Accessories	700
Keyboard	Accessories	1200
Monitor	Electronics	14000
Headphones	Accessories	2500
Printer	Office	12000
Chair	Office	6500
Desk	Office	9000

Task 9 — Pivot Summary & Chart

Category-level summary and chart are included in the workbook.

Task 9 — Pivot Summary Data

Category	Total_Sales
Accessories	33600
Electronics	428000
Office	74500

Conclusion

The package covers the complete nine-task SQL and Excel assignment with task-wise scripts, datasets, outputs, workbook, evidence, README and report.